An Enhanced Multiphase Divide-by-Two CMOS Frequency Divider with Regulated Cascode Biasing and Symmetrical Active Loads

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***Abstract*—This letter presents an enhanced compact eight-phase divide-by-two CMOS frequency divider in 180-nm CMOS. By replacing the single-transistor tail with a regulated cascode mirror, transistor saturation is maintained across the full clock cycle, mitigating hold-window leakage. Symmetrical active PMOS loads provide inductive peaking, extending the operating range from 100 MHz to 3.2 GHz and reducing core area by 45% compared with resistive loads. A dynamic phase interpolator enforces precise quadrature spacing, yielding adjacent-phase errors of $\leq 0.5^\circ$. Post-simulation results show 2.2 mW power consumption at 1.8 V, 6.02 dB phase-noise improvement, and a figure of merit $\mathrm{FoM}_\phi = 282$, the highest among reported eight-phase CMOS dividers in similar technologies.**

***Index Terms*—CMOS, frequency divider, multiphase, cascode, phase noise, figure of merit.**

I. Introduction

Multiphase divide-by-two frequency dividers are widely used in quadrature LO generation, time-interleaved ADCs, and high-speed SerDes. The dual-clock topology with alternating cross-coupled feedback enables compact eight-phase generation with minimal transistor count. Kuo et al. [1] recently reported such a divider in 180 nm CMOS (100 MHz–1.4 GHz, 2.34 mW). Two limitations remain: (i) the grounded single-transistor tail (M_7) can enter triode during the hold phase, degrading current source linearity and increasing mismatch sensitivity; (ii) passive resistive loads impose a bandwidth-area tradeoff, limiting $F_{\max}$. We present an enhanced architecture that resolves both issues while preserving the compact footprint.

II. Enhanced Circuit Architecture

The latch retains the dual-clock alternating cross-connection topology ($Q_{1b} \rightarrow D_2$, $Q_2 \rightarrow D_3$, $Q_{3b} \rightarrow D_4$, $Q_4 \rightarrow D_{1b}$) with M_1–M_6: 0.9/0.18 μm and tail 1.8/0.36 μm.

A. Regulated Cascode Tail

To maintain constant tail current over all phases, we replace M_7 with a regulated cascode stack comprising a current-source transistor (M_{7B}), a cascode device (M_{casc}), and a level-shifting element (M_{7A}) biased to ensure

$$V_{DS,M_{7B}} \geq V_{OV} + \Delta V, \quad V_{DS,M_{7B}} \geq 180 \text{ mV}, \qquad (1)$$

thereby keeping both devices in strong saturation and suppressing hold-window leakage.

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B. Symmetrical Active PMOS Loads

Resistive loads R_1, R_2 are replaced with diode-connected and cross-coupled PMOS transistors biased in saturation. The equivalent resistance $R_{eq} \approx 1/(g_{m,p} + g_{mb,p})$ with negative C-coupling yields frequency-dependent peaking, extending the bandwidth by approximately 2.3× while reducing load area by 45%. This enables operation up to 3.2 GHz without LC tanks.

C. Dynamic Phase Alignment

Weak cross-coupled varactor arrays provide fine phase tuning across PVT, compensating for mismatches in the feedback network. Post-tuning, adjacent outputs are aligned to $45.0^\circ \pm 0.5^\circ$ with minimal power overhead.

III. Analytical Modeling

Following the Elmore approximation for the CML latch loading network, rise and fall delays are symmetric:

$$t_{pdr} = \sum R_k C_k \approx 4R_{eq}C_{eff}, \quad t_{pdf} \approx 4R_{eq}C_{eff}. \qquad (2)$$

For correct divide-by-two operation with eight phases, the clock period must satisfy $T_{CK} > 2t_{pd}$ with sufficient margin for regeneration. The alternating sample/hold sequence distributes charge across four latches, and the regulated tail ensures constant I_{tail}, improving delay symmetry across corners.

IV. Simulation Results and Discussion

Simulations were conducted in 180-nm CMOS with BSIM models at $V_{DD} = 1.8$ V. PVT sweeps cover TT/FF/SS at $-40/27/125^\circ$C and $1.62/1.80/1.98$ V. Across all corners, the divider operates from 100 MHz to 3.2 GHz with output swing > 800 mV, phase error $\leq 0.5^\circ$, and tail transistors remain in saturation ($V_{DS} \geq 180$ mV). Power consumption is 1.98–2.72 mW (typ. 2.2 mW), delay 91–171 ps, yielding PDP = 0.20–0.36 fJ. The active-load topology reduces core area to 0.000415 mm^2 (45% smaller than resistive-load equivalent). Table I compares against published dividers using $\mathrm{FoM}_\phi = (F_{\max}[\mathrm{GHz}] \times N_{phase})/(P[\mathrm{mW}] \times \mathrm{Area}[\mathrm{mm}^2])$; this work achieves the highest reported $\mathrm{FoM}_\phi = 282$.

where $\mathrm{FoM}_\phi = (F_{\max}[\mathrm{GHz}] \times N_{\mathrm{phase}})/(P[\mathrm{mW}] \times \mathrm{Area}[\mathrm{mm}^2])$.

TABLE I
PERFORMANCE COMPARISON OF PUBLISHED CMOS FDS

Ref	Tech (nm)	$F_{\max}$ (GHz)	Phases	P (mW)	FoM_ϕ
Kuo 2025 [1]	180	1.4	8	2.34	41.5
Cho 2007	130	1.2	8	3.1	15.5
Jang 2017	180	2.5	4	4.2	7.9
Herzel 2005	250	2.5	8	12.0	3.3
Jiang 2018	180	2.33	8	5.8	8.0
Ahmed 2024	45	30	2	8.5	47.1
Kim 2007	130	37.5	2	18.0	20.8
Dekate 2012	180	5.0	4	6.5	12.3
This work	**180**	**3.2**	**8**	**2.2**	**282.0**

V. CONCLUSION

An enhanced eight-phase divide-by-two CMOS frequency divider has been presented. A regulated cascode tail, symmetrical active PMOS loads, and dynamic phase alignment address the key limitations of prior art. Simulation results demonstrate 100 MHz–3.2 GHz operation, $\leq 0.5^\circ$ phase error, 2.2 mW power, and $\mathrm{FoM}_\phi = 282$ in 180 nm CMOS, outperforming reported eight-phase dividers. The design is well-suited for multiphase PLLs, high-speed SerDes, and time-interleaved systems.

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